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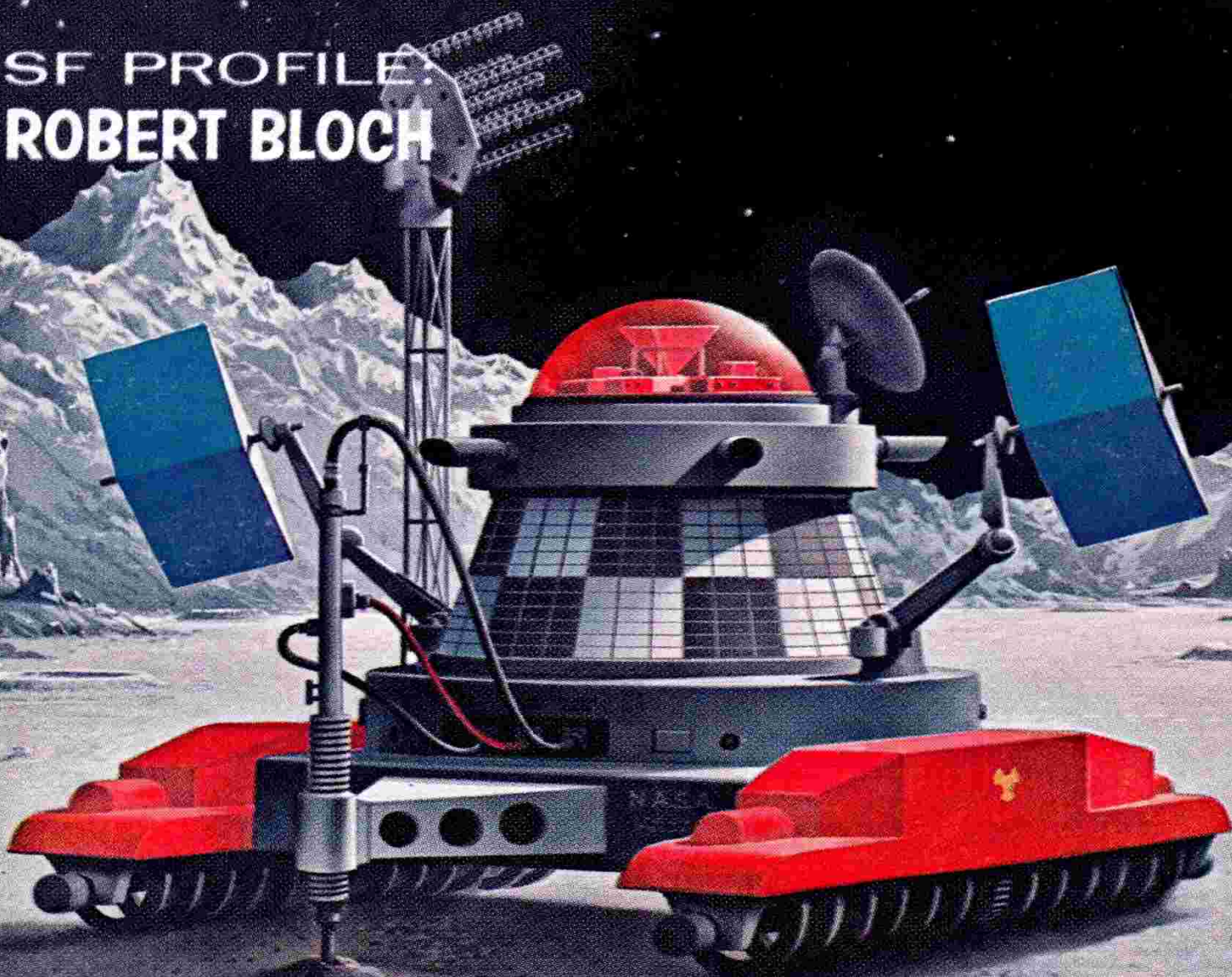
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by Raymond F. Jones

SF PROFILE:
ROBERT BLOCH



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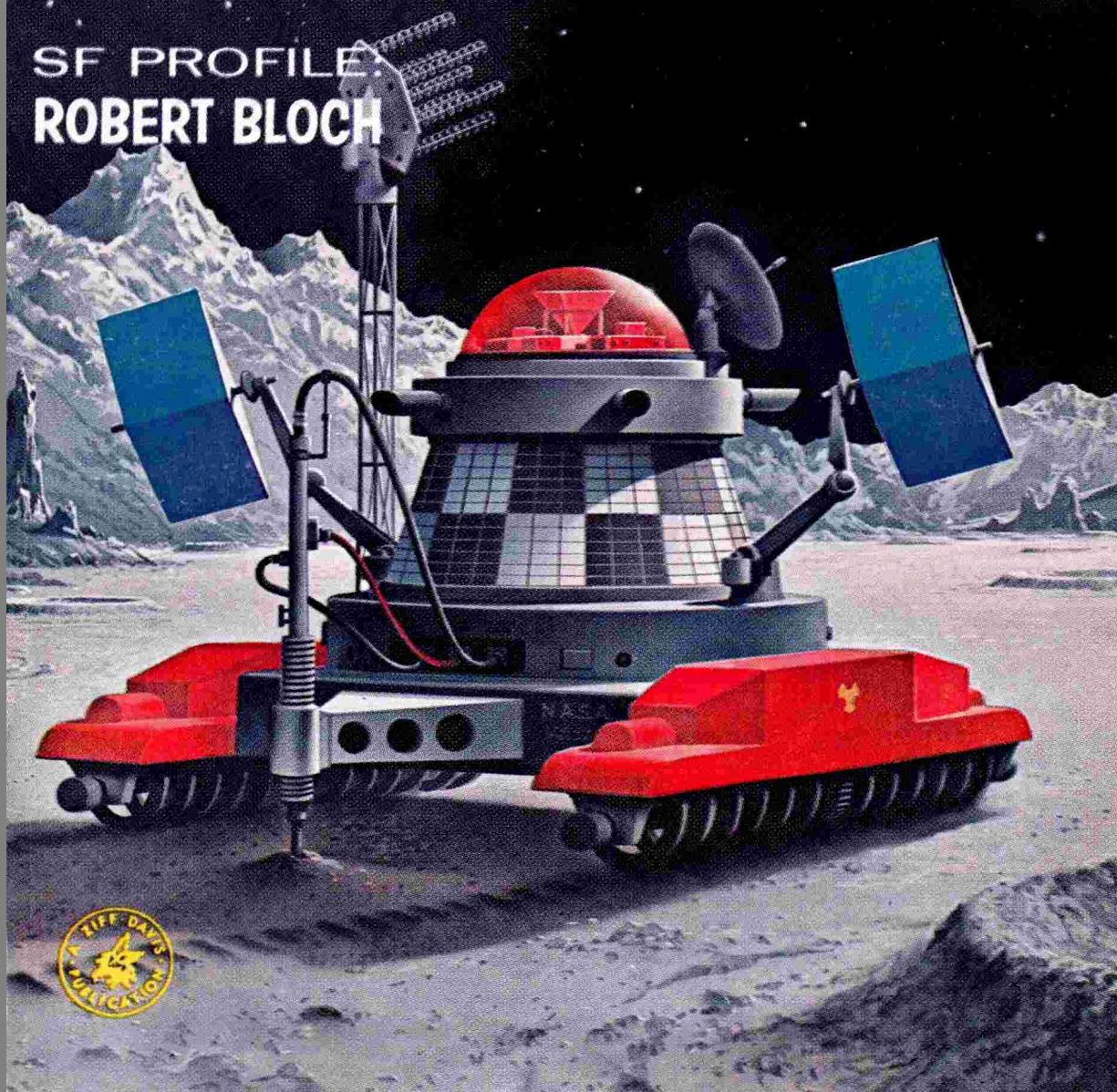
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*** START OF THE PROJECT GUTENBERG EBOOK STAY OFF THE
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Stay off the MOON!

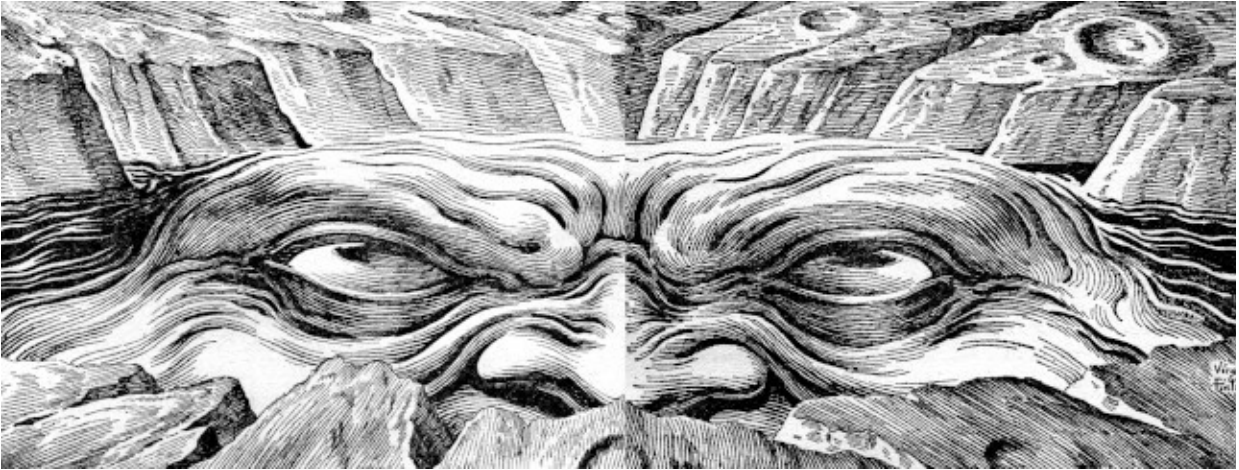
By RAYMOND F. JONES

Illustrated by FINLAY

*How do you fill a pipette and measure out
a half c.c. of hydrochloric acid into a test
tube—from a quarter million miles away?*

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The real problem, of course, is not quite that simple. You don't literally fill a pipette or use a test tube; you activate metering circuits that force tiny, ground-glass plungers a measured distance into reagent pumps. You send signals that close some valves and open others, and apply heat and adjust temperatures, and filter solutions, and send the product to a spectrometer that determines what you've got and how much.

Then you have to code it and get the information from the moon to earth.

James Cochran had seen the equipment work through hundreds of checkout analyses. But he didn't understand it. He was a chemist, and he had drawn up the specifications for the chemical analyzers of the Prospector, but it had been the electronic boys who dreamed up the remote mechanization and the telemetry equipment that would allow him to sit before a complex panel at the Center and direct his chemical laboratory on the moon to learn what the moon was made of. Some of the light-headed technicians who worked on the project had dubbed it Operation Green Cheese, but Cochran had more respect for the complexity of the effort.

It was Sunday midnight. The beginning of countdown was forty hours away. Cochran's crew had finished the chemical checkout, but in the assembly hangar technicians still swarmed about the Prospector, giving final-check to the power and telemetry components.

Jim Cochran signed off the last of the check reports and dropped them in the slot for delivery to the Project Director. He turned off the lights over his own desk and went out to the hangar. Under the blaze of fluorescent lights the device looked like some monstrous insect. The differential housings over the worm-screw drives gleamed like a red, segmented carapace. The blue appendages of the solar cell boxes were extended as if in some frantic appeal. The radar dish and the helical antenna extended mutely upward. And, like a furious proboscis, the exploratory drill, which would pierce the moon's skin to a depth of five hundred feet, seemed to gnaw at the concrete floor of the hangar.

Sam Jarvis, supervisor of electronic checkout, saw Jim Cochran enter and came over to him with a broad, weary grin. "AOK, so far! This package is going to be perfect. If only the rocket boys will set up a bird that will take it to the moon—"

"They'd better," said Jim. "I don't think I could ever go through this again if they dump the Prospector in the drink."

Sam turned back to look at the robot machine and the swarming technicians. "Yes, you could," he said. "All of us have gone through heartbreak time and again the past five years, watching them blow up, or fall back and burn in the atmosphere because the motors didn't ignite. Or seeing them get all the way to the moon and have some five-dollar transistor conk out. But we always have at it again. You will, too. You're new, but you're one of us now. You never back out when you've come this far."

Watching the Prospector, Jim knew Sam was right. It had taken some persuasion to bring him to this point, however. Until a couple of years ago he had believed he would be content with ivory-tower plastics research for the rest of his life. The persuasion had been applied when Mary's brother, Allan Wright, had made the astronaut team.

Allan and Jim had grown up together. There was no other person Jim felt closer to except Mary and their two children. Allan had dreamed of space when they were kids, and when he was fifteen, he said, "I'm going out there. I don't know how. But, somehow, I'm going out there."

Now, he had been selected to captain the first Apollo voyage. He had been born for that purpose, he said.

But while he was still in the general pool of astronauts he had opened his campaign to get Jim into the space program. "They need the best brains they can get," Allan said. "You haven't got any right to sit in a musty old plastics lab while guys with half your ability try to get us into space. NASA will take you tomorrow!"

Jim didn't try to tell him that his plastics lab wasn't exactly musty, or that he didn't think of himself as one of the best brains in the country. But Mary sided with Allan; she was almost as excited about space as he was. In the end, Jim went to NASA. Within days, he had been assigned to head the development of the Prospector chemical mechanization.

It had been something of a jolt to pull up all the roots he had so carefully put down for him and his family, and move to the hectic, bustling, space-frontier community of the Center. But he wasn't sorry. It put something new in the blood, something men had never known before.

Space!

The great Saturn lifted slowly, on a vast blossom of fire, with snowy lox streaming down its sides. Then it was gone, a twinkle of fire high above, among the stars. That was all.

Mary and Jim Cochran continued to stare at the fading twinkle, and finally they turned away. Allan had obtained permission to be in the blockhouse during the firing. It hadn't been necessary for Jim to be there. He didn't want to know the instant-by-instant telemetry reports which told whether or not the flight was successful. Sam or Allan would call him when they knew. That would be soon enough for him.

"Let's drive down to the beach and watch the moon from there," said Mary. "We can't just turn around and go home, like—like nothing had happened."

Jim smiled in the darkness. Mary was as eager as he was for the success of the flight. And she didn't have his fear of failure, that kept him from wanting to know the maybe-yes, maybe-no indications that the telemetry would first show.

"Sure," he said, "that's a good place to watch it."

The moon.

They watched its reflection thrashing on the water as the breakers rolled across, under, and around it. It was the same image that men had watched and wondered about and feared—for a half million years. The first creatures that had any semblance of manhood had sat on their haunches on this same shore and watched the same moon and the same water.

And felt the same fears, Jim thought.

He didn't know whether it was fear or not, but there was some sense of awesome mystery that filled him when he looked at the moon. It had been that way all his life. He remembered how it was when he was a boy and he walked through the fields at night on his way home. He had to pass Cramer's Pond, and when the moon was up its light from the sky and its reflection from the pond seemed to fill the whole earth with a cold, silver light. He always hurried past the pond on such nights.

Mary felt it, too. "I wonder why the moon makes people feel the way they do."

"How does it make people feel?"

"Oh, kind of—kind of—*you* know!"

Jim laughed aloud. This was a typical Mary Cochran explanation. But it told him all he needed to know. What she said was quite true. He *did* know.

The baying of dogs on a wintery, moonlit night.

The madness called lunacy.

Seeds must be planted just so, in relation to the moon, or the crop will fail.

Men had always felt strange things about the moon. Would a Saturn missile and a mechanical monster in its nose be able to destroy all that?

Jim started the car. "Let's get back. I've *got* to know how the flight's going!"

They were still in the blockhouse, but the tension was relaxed. They were talking and watching the meters and cathode ray tubes without the strain and fear of failure. Jim knew the answer even before Sam and Allan walked up to him and slapped him on the back.

"Where the devil did you go?" said Sam. "I thought you were going to be right behind me when we fired, and you weren't here at all!"

"It's like your first baby, you want to be there, and you don't. Was it a good shot?"

"Was it a good shot?" Allan's face became ecstatic. "We've never had a better one. On course all the way!"

The Project Director, Emil Hennesey was behind them. His face was bleak. "I expected you to be here for the firing, Cochran," he said. "It seems to display little interest in your end of the project that you didn't feel it necessary to show up."

Jim looked at him steadily and shrugged without answer. Hennesey was one guy whose presence on the space team Jim couldn't figure out. He was an ex-Major, and he had no capacity for dreaming. Men, machines, transistors, rockets—they were all the same thing, merely objects to be made to obey.

"You are aware of your next sequence of duties, I trust?" said Hennesey.

Jim nodded curtly. "I'll be ready."

Sixty-six hours to the moon. That's what it takes with marginal escape velocity and free-fall conditions. But it was really five hundred thousand years and sixty-six hours, Jim thought. Surely there hadn't been a single hour in all that time when someone, somewhere on earth had not felt the longing to solve the secret mystery of the moon.

Now they were about to find the answer. But what would they have when they found it? They would know that the surface dust of the moon consisted of certain percentages of silicates and oxides. They would know that the under layers were composed of rocks, maybe of granite or limestone or basalt. They would determine how much of each.

And then it would be over. The quest of the ages would be answered with a few simple statements that could be obtained in any high-school chemistry lab—if the lab were on the moon.

Jim Cochran felt there *had* to be more to it than that.

Why do dogs howl at the moon on winter nights?

Why do men say that madness of the mind is lunacy?

Why must planting be done in the right phase of the moon?

Little sleep was had by any of the crew during the next two nights, even though the instrument stage of the ship was now completely inert except for occasional telemetry signals that were fed to the computers for course checking and correction. The ship was simply falling on its own momentum.

Six hours before moonfall, activities in the tracking center accelerated and the tension increased. There was no question of hitting the moon; the landing had to be made safe for the cargo of instruments.

Jim Cochran watched the operators during this period. He told himself he didn't understand it, but he had actually learned a great deal of electronics during the past two years. He had had to in order to design and operate a chemical laboratory 240,000 miles away.

The television screens came on, showing the pock-marked surface of the moon as the ship orbited. The thrill and the fear of the great unknown began to rise in Jim's throat. By the silence in the room, he was sure the others sensed it, too.

Abruptly, the braking command was given and the ship began to fall out of orbit towards the planned landing in the Sea of Rains. On the screens, the images swelled as the ship plummeted faster. In one corner could be seen the spring-loaded extension legs, like those of some great spider. It seemed impossible that these could cushion the violent shock of landing.

The sudden surge of a retro rocket and the blast of moondust blinded the television eye, but there was a sense of crazy, rocking, rolling motion. Then the eye went dead.

Jim almost cried out. The ship couldn't have crashed.

An operator quickly switched controls and the screens came alive again. He turned a dial slowly. The camera eye moved. It swept the craggy horizon and the nearby floor of the Sea of Rains. Others had seen this before, but it was the first time for Jim. He found himself pushing forward, drinking in the sight eagerly.

"The moon—the moon—" he said softly to himself. But the others heard it and they understood.

Signals were sent across space to collapse the landing legs and unfold the sides of the instrument cone like the leaves of a flower. The Prospector lay exposed to the environment for which it was built. Slowly, in response to other signals, the worm-screw drives, which had been retracted against the body of the vehicle, turned through an arc and lowered to the surface. Locked in position, the drive screws began turning slowly. The vehicle moved off the now-useless landing support and became an entity of its own.

The ungainly arms of the solar cells automatically oriented toward the sun; the antennas pointed toward earth. The scanning cameras in the turret of the Prospector took control of the video circuits and the turret slowly turned as the vehicle moved across the face of the moon. The landing support remained behind and slowly dwindled like some useless wreckage.

There was sudden pandemonium in the tracking center control room. The operators laid down their headsets and began pounding each other on the back, while ear-splitting Indian yells filled the air. Jim and Sam found themselves beating each other on the arms and yelling senselessly.

"We made it!" Sam cried. "We made it! We got your little old laboratory up there for you!"

There were hours of testing and calibration yet to be done before the Prospector could be used for its primary mission. Hundreds of electronic circuits had to be checked to see that they survived the takeoff and landing without becoming distorted or inoperative.

Jim went home for the rest of the night. When he returned the next morning Sam reported that all circuits were go, and the Prospector was his.

He had operated the laboratory in the Prospector many times, either on a mock-up or from this control panel while the Prospector was in the hangar. But he couldn't keep the faint tremor from his hand as he reached for the first control that would manipulate the machine on the moon.

The drill had been extended to operating position, but the head had not yet been energized. Jim touched it to the fine dust of the floor of the Sea of Rains. The drill went quickly to a depth of eighteen inches in the dust before it struck something firmer.

"That kills the theory about eighty feet of that stuff, anyway," said Sam as he read the instruments.

Jim was not interested in depth at this time. He fed some of the surface material into the laboratory and set the controls to run the preprogrammed analysis. They waited minutes; then the analysis began to appear in cryptic symbols on a paper tape.

Jim glanced at it and frowned.

"What's the matter? Isn't it working right?" Sam asked anxiously.

Jim hesitated. "It indicates the presence of several silicates, some carbonates, and a high percentage of oxides. These are mostly of sodium, calcium, and iron, as you might expect. But there's something wrong with your calibration. The atomic and molecular characteristics aren't coming through right."

"The boys ran checks on the standard samples aboard the Prospector last night," said Sam. "The results tallied exactly. I'll show you the tapes."

Jim waited, puzzled, while Sam brought up the check tapes. When he saw them, he shook his head. "There's a standard calcium carbonate sample carried aboard the Prospector. Here's a calcium carbonate picked off the surface. You can see the difference yourself. The nominal analysis is the same, but the atomic weights and the energy levels are just slightly different. That doesn't make sense unless your circuits are out of calibration."

"Let's run another standard sample," said Sam.

Within a few minutes the calibration check had been repeated. Jim held up the tape. Sam peered over his shoulder. "Just like the first one," said Sam. "Nothing's wrong with the circuits. Maybe you've got some new stuff there, that's never been identified before."

"That's hardly possible," said Jim. "There aren't any new elements in the places where sodium and calcium and silica are supposed to be. Yet, I don't understand how this can be. If the atomic weights are different, and the energy levels are different, they have to be different elements. It doesn't make sense."

"Well, why don't we push on," said Sam, "that is, if you've completed the surface sampling in this spot. Some samples at lower depths may give other indications."

Jim agreed. He drove the drill deeper into the face of the moon. At ten-foot intervals he removed samples and ran them through the analyzer. The results were the same down to the hundred-foot level. All results showed common chemical elements with slightly variant atomic characteristics.

Which made them *different* chemical elements!

After six hours, Jim stood up from the console and shook his head wearily. "It's no good, Sam. There's something wrong that I can't put my finger on. If it isn't in the circuits, I don't know where it is. But these readings just aren't right. There's no use going deeper until we find out where the error is."

Sam's face was somber. "There just isn't any error. There can't be. Unless it was made by whoever put the moon together—"

"Please make a complete check of every analyzer and telemetry circuit tonight, and we'll try again tomorrow. I want to think about this."

He thought about it, and he dreamed about it. And along about three o'clock in the morning he sat bolt upright in bed and stared at the dim moonlight on the opposite wall of the bedroom.

It wasn't possible, he told himself audibly. It just wasn't possible!

Mary stirred and leaned on one elbow beside him. "What's the matter. Are you having nightmares?"

"Yeah—yeah, I guess I am. I'll be back in a minute, honey." He got up and padded to the door. "I've got to make a phone call."

"At this time of night?"

But Jim was gone. He turned on the hall light and dialed Sam's number. After a long time Sam answered sleepily.

"Wake up!" said Jim. "I've just figured it out!"

"Who the devil—? Oh, it's you, Jim. Figured out what? Do you know what time it is."

"Do you know the results of your calibration re-check?"

"How would I know that? I've got the night crew on it, but I didn't ask them to report to me in the middle of the night. Go back to bed, and let's talk about it in the morning."

"They're not going to find anything wrong, Sam."

"I could have told you that."

"But the elements of the moon are different—and there's only one explanation."

"What?"

"Think about it a minute, Sam. We take a spectrograph of the sun, and we find the same elements that are here on earth. We turn it on Alpha Centauri and find the same thing. We turn it to the farthest stars we can find that give enough light to record by. Always the same. Calcium is calcium, whether it's on the earth or on a star a half billion light years away."

"So?" Sam's voice was tired, and he sounded as if he was listening only because Jim was too good a friend to tell to go to hell for calling in the middle of the night.

"So? So what?" Sam repeated.

"So we go to the moon," said Jim, "and all of a sudden calcium isn't calcium, and the sodium on the moon isn't the same as the sodium on earth and on the

sun and on Alpha Centauri and the stars a half billion light years away. Don't you see what that means!"

"No, I guess not," said Sam dully. "Maybe in the morning—"

"It means the moon just doesn't *belong*, Sam! It means the moon is completely foreign to anything in the Solar System, in the whole galaxy—in any galaxy we have been able to analyze. It means the moon has come from somewhere else, from a region of space where atoms and electrons are not even the same as atoms and electrons here. It must be a somewhere that's so far away it's beyond the edge of space as we know it!"

"I'll get dressed and come over," said Sam.

Mary made chocolate and toast, and they sat around the kitchen table thinking and talking of the awesome implications of Jim's theory.

"If what you say is true," said Sam, "it might be that the slightest contact with any substance of the moon would be sheer poison to a human being. A returning vessel could never be permitted to enter the earth's atmosphere, and decontamination would become one of the major branches of science."

"That's entirely possible. It would complicate enormously the problems of establishing a moon-base. A speck of moondust inside the base might be as lethal as an unshielded reactor."

Mary was looking out the kitchen window toward the thin crescent of moon that was setting over the city. She thought of Allan, who would soon be voyaging to that alien world. "It's like a trap up there in the sky. We should never have tried to reach it."

"No—it's not like that at all," said Jim vigorously. "We'll solve whatever problems we find there. But think of it! We don't have to build a ship capable of crossing billions of light years of space to see what's out there. Something from out there has come to us and parked right in our own front yard. We have a thousand times more reason to go now!"

Sam toyed with his toast and dunked it in the chocolate. "I think you ought to keep it quiet—until we really *know*, don't you."

"Why? I'll run some more tests, sure. I'll plug every loophole there can possibly be. But unless I find something new I'm going to announce it. Why shouldn't I?"

"I don't think Hennesey will like it, for one thing. Too sensational. Even when we actually land there and confirm your analyses by on-the-spot checks—it's still only a theory that the moon doesn't belong to this galaxy. You'll never be able to prove that."

"What we've found already is proof enough!"

"Not for Hennesey. He'll ask to see the shipping manifest by which the moon was transferred here. You know Hennesey."

"Sure, I know Hennesey," said Jim bitterly. "And he doesn't count. Not in something like this. This is big, and important, and I'm going to announce it. I want the credit for discovering it. It'll be called Cochran's Theory, and some University will offer me at least an honorary Ph.D. Is that bad?"

Sam shook his head. "Of course it's not bad. But I wonder what the public reaction will be like, and how about Congress—especially if this business about possible poisoning from moondust turns out to be correct? There might be a lot of pressure to cut funds and maybe cancel the whole moon project."

"If moondust is lethal, there's no better time to be warned against it than right now!"

He spent the following week, eighteen hours a day, at the chemical analysis panel in the tracking station. As long as the moon was above the horizon, day or night, he kept experiments going—checking, re-checking, calibrating, searching for loopholes.

There were no loopholes. There was no malfunction of equipment. The atoms of the moon's elements were not the same as those of the rest of the galaxy.

When Allan next came to visit on leave from his own station, Jim told him what had been found. Allan's face paled a little as he listened in awe to the

story. He stood up and paced across the room as Jim finished.

"I always knew there would be something—something extra in this," he said. "You look up at night and you know the moon can't be just another piece of earth. Now—you've found out what it is!

"How far could it have come? How old can it be? Imagine stepping out onto a world so old, from so far away!"

"You haven't heard it all," said Mary tensely.

"There's more?"

"They think perhaps the slightest touch of moon substance may be lethal."

"We don't know," said Jim hastily. "It's a thought that Sam advanced. But I think it's quite possible. We won't know until we can run animal tests. But special precautions will have to be taken to decontaminate the Apollo after you re-enter with your space-suits."

"This will mean engineering changes. Are they under way?"

"I haven't announced my theory yet. But the necessary changes to provide for decontamination will have to be made."

"I wanted this as much as you," said Mary. "But now it's gone all wrong. We should have gone on dreaming about the moon and let it be a dream that never came true. I'm afraid of it now. No man should set foot on something so alien and so different."

Allan put an arm around her shoulders and shook her gently. "Sis! What kind of talk is that? You're talking to the guy who's going to be the first one to put his foot on the moon."

From the mountain of data accumulated in his experiments, Jim wrote his paper. Hennesey could control the publication of any material based on space experiments to the extent that they affected national security. But even he could not find a means to extend a security blanket to cover a theory of extra-galactic origin of the moon.

He raged at Jim, however. "You'll make a laughing stock of us in every scientific center of the world! You can't publish a ridiculous thing like this!"

"No one will laugh if he reads the data I've got to present," said Jim. "Every member of our staff who knows the subject has verified that the data are correct. The conclusion is inescapable."

"I can't forbid publication, Cochran," said Hennesey, "but I think it is very unwise for you to go ahead. Very unwise."

"I'll take that risk," said Jim.

He sent the paper to the Journal of Astro-physics. At the same time he sent an announcement to the major news services.

He had expected some sensationalism in the reporting. There was more than he bargained for. Some of the headlines that followed were:

"Savant Says Moon is Messenger from Outer Space."

"Moon Will Poison Earth."

"Moon Trip—One Way only."

The reactions in the upper echelons of NASA were almost as bad—in their own way. No thought had ever been given to a need for complete decontamination of astronauts and equipment after exposure on the moon. The requirement, if admitted, would threaten the entire program in the minds of some of the engineers. Others admitted it was tough, but thought they could solve it in an extra year or so. Rumblings were heard echoing down from Congressional halls. Why hadn't the stupid scientists known in the beginning that this was necessary? Always bungling things—

In the end, it was Alan himself who came up with a proposal that kept the project from bogging down and still provided some measure of protection against the possible menace. He suggested a plastic outer suit to be fitted over the space-suit and discarded as the astronaut re-entered the space vehicle. With care, such a procedure could prevent direct contact with moondust. In the meantime, it was hoped that robot vehicles could bring back moon samples before the Apollo was sent out.

This rather mild proposal did much to calm the furor in NASA and contractor engineering circles and soon the press had abandoned it for other, more sensational stories. But Hennesey and a number of other officials didn't

forget. Some of them believed Jim Cochran was a charlatan at worse and an incompetent at best. They considered he had degraded American science with his fantastic theory.

Scientific judgement was being held in abeyance until actual moon samples were available on earth. For the present, at least one of Jim's predictions had come true. The hypothesis was becoming known as Cochran's Theory. That it was also called Cochran's Idiocy by a few didn't matter.

Jim continued his own sixteen-hour stints at the analyzer controls, probing in a wide pattern over the floor of the Sea of Rains, and striking deeper toward the heart of the moon with each probe.

Probing to such great depths was made possible by a development that didn't even exist when the Prospector design was begun. Then, it was hoped that penetration to a foot and a half of the moon's surface might be possible. Five-hundred-foot holes were only a madman's nightmare. How could you carry such drilling equipment all the way to the moon?

Then, in the last months of Prospector design, laser devices had been produced, capable of burning holes in a diamond. It was only a small step, then, to the design of a drilling head which mounted a cluster of laser beams. These would literally burn their way toward the heart of the moon.

The laser drilling head was lowered on five hundred feet of minute cable, which had tremendous tensile strength. The vaporized moon substance boiled out of the hole and condensed above the surface, settling as fine dust. As the hole deepened, the condensation products coated the upper portions of the hole and the cable. To keep the hole from thus being closed, the cable was vibrated at a frequency that shook loose the condensing rock products, and the laser head was raised with beams shooting upward to clear the hole.

Jim found that a very special technique was required to raise and lower the head at the proper intervals to keep the hole clear and prevent loss of the drilling head. A spare was carried, but he didn't want to face the loss of even one. After three weeks, he felt confident in his operation and began lowering the drilling head to depths of two hundred and three hundred feet.

As he had expected, along with the lunar geologists who were participating, the moon showed a definite pattern of stratification. But the differences between the layers seemed slight. Chalky, calcium compounds were abundant. Some were powdery; others were pressed into brittle limestone formations. No really hard rocks such as granite were encountered, however. The boundaries between layers were ill-defined. No one knew what to make of it. The observations were interesting. Explanations were wholly lacking.

Then, after five weeks of probing, on the edge of the four-hundred-foot level, Jim found something new. He sought out Sam at the end of the day.

"A few years ago," he said, "scientists were startled to find chemicals that were the product of life, inside meteors from outer space."

"I understand they've even found bacteria which they have been able to bring to life," said Sam.

Jim nodded. "More than four hundred feet deep on the moon I've found the same kind of chemicals—hydrocarbons that must be the product of living cells."

"Four hundred feet deep on the moon—" said Sam musingly. "And maybe the moon came from billions of billions of light years across space. So wherever it came from there was something living. What is it? Traces of bacteria, or chemical remains of plant life like our coal mines?"

Jim shook his head. "I don't know yet. I'm not sure we can find out until we go there. But, as you say, it means the moon was once the scene of life—wherever it came from."

"One thing I haven't understood," said Sam, "is why the moon stopped here if it had been traveling through space for so long. Why didn't it keep on going?"

"It was just a combination of factors," said Jim. "The moon happened to be traveling at just the right speed. The earth was in just the right place at the right time. As a result, the moon fell into an orbit around the earth. Pure accident."

"A lucky accident!" said Sam.

Jim looked up at the pale moon above their heads as they walked toward the parking lot. "I hope so," he said. "We will soon know whether it was a lucky

or an unlucky accident."

The moon laboratory had not been designed for extensive organic chemical analysis. There were only a few things it could do with organic compounds. But these were sufficient to convince Jim that the moon had once been the scene of life.

Why so deep? he wondered. Nothing had been found in the upper levels, unless he had missed it—he would have to check that out later.

As the drilling head moved slowly downward, the evidence of fossil hydrocarbons increased. There seemed to be an almost geometric increase in concentration after he passed the four-hundred-foot level. He was certain the drill was penetrating a bed of fossil remains of some form of life that flourished the little planet that the moon must have been incalculable eons ago.

The more he thought about his theory, however, the more difficult it became to explain all the factors. If the moon had actually been a planet of some far distant system, what had torn it loose from its parent sun and sent it careening through space? Had its sun exploded, blasting whatever planets the system held into the depths of space? Such an occurrence might explain the sterility of the moon's surface, but why was the evidence of life buried so deep? Perhaps the upper layers of the moon's surface consisted of debris blasted from the exploding sun. Such debris would have been molten, flowing about the moon's surface, cremating everything living. Finally, it would have shrunk in the cold depths of space and wrinkled into the vast mountains and cracks that laced the moon's surface.

It was one way it could have happened, but it seemed so fantastic that Jim had difficulty in convincing himself that it was true.

He doubted the accuracy of his analyses. There were so many tenuous links between the substance on the moon and his own senses that an error in any one of them could destroy the accuracy of the results. But he had no reason to doubt.

He began making calibration checks before and after every analysis. It added scores of hours to his work. Sam sat beside him, checking and verifying the accuracy of the telemetering circuits constantly. The operation was as foolproof as their science could make it.

"You've got to believe what you find," said Sam. "There's no other answer."

And then, one day, Jim found an answer that was utterly impossible to believe. His mind balked and closed up completely at the thought.

Sam had been watching him for almost three hours, aware that something had perturbed Jim exceedingly. Sam kept his mouth shut and leaned quietly against the desk of his own console, keeping check on the circuits while he watched Jim grow more and more distressed. Sam didn't understand the processes, but he was aware that Jim had been going over and over the same analysis for almost two hours. At last Jim's face seemed to go utterly white, and his hands became motionless on the console.

Sam waited a long time. Then he asked, "What is it, Jim? What's the matter?"

Jim continued to stare at the panels of the console, then answered as if from some far nightmare distance. "Two chemicals, Sam," he said. "One of them a big molecule, something like hemoglobin. And neither of them could exist as fossils. Their structure would have broken down long ago. They could exist only in *live* tissue!"

He continued staring. Neither of them moved. Sam felt as if he had just heard something in a nightmare and had only to wait a minute until he woke up. Then it would be gone.

Jim turned his head at last and faced Sam. He gave a short, harsh bark of a laugh that sounded half-hysterical.

"We'd be off our rockers, wouldn't we Sam? Clear off our rockers to believe there could be something alive five hundred feet inside the moon!"

"Sure—and if it were alive, it wouldn't be sitting still while the laser beams drilled a hole into it. Besides, we just couldn't be lucky enough to lower the drill right smack into some cave where a moon bear was hibernating. All the circuits must have busted down at the same time. We'll fix it tomorrow. Let's get the girls and have a night on the town."

It was a very unsuccessful night on the town. Jim and Mary, and Sam and his wife went to a show and a nightclub.

"You're moving like a zombie. What's the matter?" said Mary as she and Jim danced together.

"Feel like a zombie. Why don't we give it up and go home? I want to get down to the lab by five in the morning."

"That's the trouble. You've done nothing but live in the lab since the Prospector landed. So we're not going home. Sam and Alice are having a good time. You dance with Alice next, and make her think you're enjoying it!"

So Jim didn't go to bed at all, but he *was* at the lab by five in the morning. The night crew were still at work. He had steered them away from the analyses he was doing so they were unaware of the shattering results he had found.

He took over the controls, and resumed work alone.

There was no doubt about it. If any of the methods they were using were accurate, then he had discovered almost indisputable proof that some living tissue existed five hundred feet below the surface of the moon.

Since the laser drilling head sealed the walls of the hole with a coating of frozen lava, it was necessary to probe horizontally for samples. Small extension drills, capable of reaching five feet on either side of the hole, were carried in the head for this purpose.

Jim lowered the head through the last twenty feet of its drilling limit. Every six inches he sent the horizontal probes to their limits. The tell-tale chemicals existed at every point. He computed the volume he had probed, and felt numb.

By the time Sam had shown up, Jim had withdrawn the probe to the surface and was moving the Prospector slowly across the moon's surface.

Sam saw the motion on the television screen. "Where are you going? I thought we were going to check out the hole we were in."

"It's been checked," said Jim. He hesitated. His original plan had been to move the Prospector a distance of fifty feet and probe again to the five-hundred-foot level. Then, decisively, he pressed the control that kept the Prospector moving. He stopped it a hundred feet from the previous hole and began the long, tedious job of drilling again to the limits of the Prospector's equipment.

Sam spelled him off during the day. By evening, they had hit the four-hundred-and-fifty-foot level. Jim took his first analysis in this hole. The chemicals were there. In greater concentration than at the same level in the previous hole.

Jim turned to Sam. "We have circuits for measuring potential differences on the lunar landscape. Could we make a reading at the bottom of this hole?"

Sam considered. "It'll take some doing, but I think we can manage it. What do you expect to find from that?"

Jim didn't dare tell him what was in his mind. "I don't know," he said. "But it might be worth trying—if there *is* anything living down there—"

By the following afternoon, Sam had made the necessary equipment arrangements so that potential readings could be obtained in the mass from which the chemical samples were being removed. The telemetered report was connected to a recorder that plotted the variations against a time scale.

As soon as the circuit was set up and calibrated, the recording meter showed a response. A very slow, rhythmic pulsation showed in the inked line on the paper.

Jim felt as if his breathing must have stopped for an infinite length of time. "That's what I thought we'd find," he said at last.

"What?" said Sam. "I don't understand what you're talking about. What do you think those pulsations mean?"

"Did you ever hear of an electroencephalograph?" said Jim, gravely.

"Electro—Sure, brain wave recordings. Jim! You don't think these waves —!"

In silence, the two men stared at the wavering pen and the sheet of recording paper that slowly unrolled beneath it.

Dr. Thomas Banning had been a class mate of Jim Cochran when they were both in their first couple of years of college. Banning had gone on into medicine, specializing in brain studies, while Jim had turned to chemistry. The two had been out of touch for several years.

Tom Banning was the first one Jim thought of, not only because of their old friendship, but because he had read recent papers describing some of Tom's new work on the frontier of electroencephalography. He called first on the phone, then arranged for a personal visit. Sam went with him. They had closed down all Prospector work while they were to be away.

Tom met them and was introduced to Sam as he ushered them into his own modest laboratory. "This isn't the plush sort of surroundings you've become used to," he said as he showed them around. "The Government isn't spending billions these days trying to find out how the human mind works."

Jim could well understand Tom's bitterness. Doing research on the frontiers of the mind, he was forced to spend his own money for much of his laboratory equipment.

"I can sympathize, but that's about all," said Jim. "I just work here myself."

"Tell me about your problem. On the phone, that sounded interesting enough to make a man's day brighter. You said something about an unknown life form with electrical pulses that might be related to brain waves?"

Jim nodded. "That's the way it looks to me."

"But where does this life form exist? Surely it can be identified!"

"If I told you, you'd throw me out or call the paddy wagon. Look at these, first."

Jim and Sam spread out the long folds of chart they had accumulated through days of recording. "Does it look like anything to you?" asked Jim.

Tom Banning frowned. "Well, it certainly *could* be an EEG record of some kind. The apparatus—"

"The apparatus was nothing but a single electrical probe, and the signal was transmitted under very unsatisfactory conditions."

"Signal transmitted, you say? Just where did this come from, Jim? You didn't come all this way just to pull my leg."

"No," said Jim wearily. "If anybody's leg is being pulled, it's mine. I wanted to see if you could recognize it as having any similarity to an EEG. Then I wanted to ask about your work you reported in your last paper. The one on 'EEG as a Brain Stimulus and Communication Medium'."

"Yes? What did you want to know about that?"

"You've had some success in taking the EEG waves of one person and applying them to the brain of another person so that the latter understood some of the thoughts of the first person while being stimulated by his brain waves."

"Yes."

"Would it be possible to do that with this record?"

Tom studied the record silently. "Any cyclic electric impulse can be applied as a stimulus to the brain. Certainly, this one can. My question still remains, however, what kind of a creature generated these pulses? If it is so alien you can't even identify it, we can't really be sure that these are brain waves. I can only say they may be."

"That's good enough for me," said Jim. "How about setting it up so that we can see if these tell us anything."

"I think I ought to make you tell me where you got these, first."

"Afterwards, please, Tom."

It took the rest of the day to transcribe the record to the format required by Tom's light-intensity reader. They set the following day for the experiment.

Both Sam and Jim were to participate. Tom applied eight electrodes to the skull of each man. They reclined in deep sleep-back chairs, and Tom suggested they close their eyes.

Jim began to feel a sense of apprehension as he heard the first faint whine of the equipment. He knew the transcribed tape was unreeling slowly beneath the photo-electric scanner. The resulting fluctuating current was being amplified, filtered, gated to the proper level, and applied to the electrodes on his skull. He felt nothing.

"Just like a ride on the merry-go-round," he said in disappointment.

Then it struck.

Like a fearful, billowing blackness rising out of the depths of Hell itself, it washed over him. It sucked at his very soul, corroding, destroying, a wind of darkness where the very concept of light was unknown.

He was not conscious of his screaming until he heard his own dying voice and grew slowly aware of the sudden rawness of his throat. He heard another screaming and it sounded like Sam. Dimly, he wondered what had happened to Sam.

Tom was bending over him, patting his face with a cold towel and murmuring, "Wake up, Jim! You're all right now. You're all right."

He opened his eyes and saw Tom, white-faced. He turned and looked at Sam, whose head lolled sluggishly while a low whimpering came from his lips.

"I'm all right," said Jim weakly. "Take care of Sam."

Exhausted, he leaned back and closed his eyes another moment. Sweat oozed from every pore of his skin, cold, fear-inspired sweat.

An hour later, he felt completely recovered from the experience, except that his knees were still a little wobbly when he tried his legs.

"We've got to try it again," Jim said. "Can you cut down the intensity a little? Better still, how about rigging up an intensity control that we can operate for ourselves?"

"Nobody is trying that thing again in my lab," said Tom Banning. "Do you think I want a couple of corpses on my hands? Not to mention the droves of police that your screaming will bring down."

"We've got to *know*," Jim said. "Listen, Tom, I'll tell you where we got this record. Then you can judge for yourself."

Rapidly, he told Tom all that had happened since their first experience with the Prospector. The brain specialist listened impassively until the end of the story.

"So you conclude there's something monstrous on the moon, and this experience you've just had would indicate that it's highly inimical to human life," said Tom.

"That's about it," said Jim.

"What do you expect to do about it?"

"I want to finish what we started here. Then I've got to show the authorities that the moon project has got to stop. We can't go ahead with our moon landings now. If we do, that thing will be stirred out of dormancy into life—and, somehow, it will make its way to earth. I wouldn't be surprised if it could navigate space alone, its own naked being."

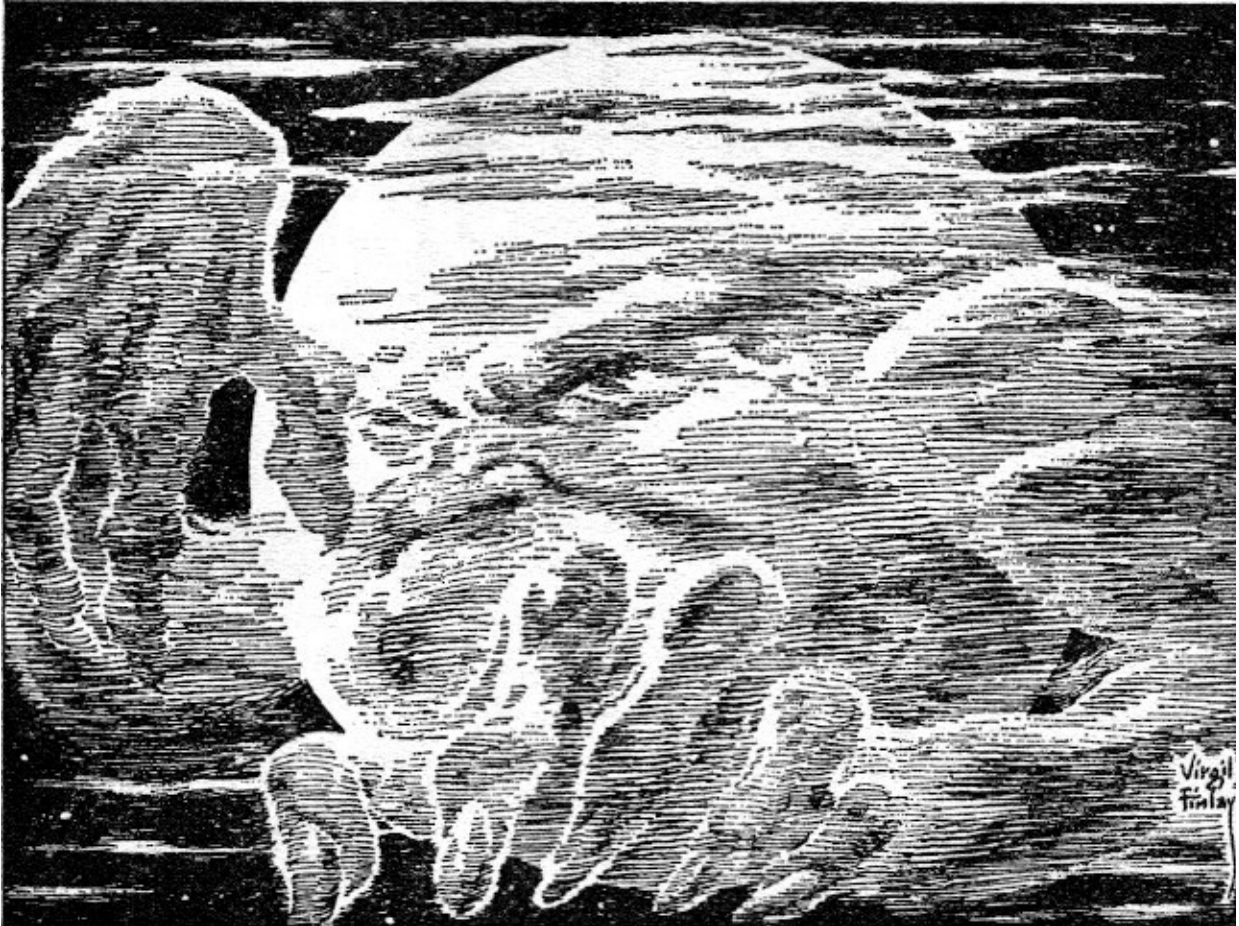
Tom turned back to his equipment. "All right, let's go. I want to get a sampling of that before we're through, too."

With a control that he could operate himself, Jim found it endurable. With the control at minimum intensity, he tensed for that first terrible impact of the alien impulses pouring into his own mind.

They were weaker, but still he felt as if the shroud of death had settled over him. He heard a moan from Sam and knew his companion was experiencing the same sensations.

The impulses of evil poured on through the electrodes into his mind. He sensed the immensity and purpose of the thing that had generated them. He sensed that out of some far reach of space, where time and dimension were

not the same, the thing had acquired an eternal nature of a kind that knew no birth and could experience no death in the dimensions of man.



He sensed that its nature and its purpose were pure destruction. Destruction of life in any form. It was a thing of death, and life and it could not exist in the same universe.

He sensed how it had come and why it had come, and the partial defeat that had sent it into dormancy because there was no life of the kind it knew in the universe through which it hurtled.

Now—it was once again aware of life.

The three of them went back to the tracking station laboratory together. Jim managed to obtain a clearance for Tom to see what they were doing. "I want to move the Prospector a long distance and try one more hole," he told Sam.

"What do you mean by a long distance?"

"A hundred miles."

"A hun—! You think you'll still find this thing that far away?"

"We'll find out. Can the Prospector travel that far?"

"Sure. If you wait long enough. Its maximum speed is two miles an hour."

"A little better than two days. Let's pick the direction of the flattest and lowest terrain. I don't want to get it up into the mountains."

During the following two days, Jim considered what his next move should be. He had to present his data and evidence to a conference of men who mattered, who could make the necessary decisions. It had to be brought to the attention of the top levels of NASA. The Department of Defense and the Presidential advisors should be in on it, too.

His thoughts came to a stop and he felt more than a little hysterical. Who was he? A third-string chemical researcher on one of dozens of current NASA projects. Who was going to let him call a conference of the nation's brass and instruct them to close down the moon program?

Nobody.

In the Civil Service hierarchy to which he belonged there was absolutely no way on earth by which he could bring his story to the attention of the people who could act on it.

No way at all. But he had to try.

He tried to reach the Director of NASA. The Director's secretary told Jim the Director was out of town and could not be reached except for emergency or other top-priority communications. Jim said that was exactly the nature of his message. The Secretary told him to get his Project Director to approve the message and an effort could be made to get it through.

That meant Hennesey.

Hennesey laughed in his face, and told him that one more fantasy like that would get him fired.

Jim had known that's the way it would be, but he had to try.

By this time, the Prospector had traveled more than ninety miles from the last probe. It was far enough, Jim decided. They'd put down one more probe, then—he didn't know where he'd go from there.

Sam saw the bleakness and bitterness on his face when he came into the tracking station. "No luck?" said Sam.

"What do you think? Have you ever realized that there is no way whatever for the ordinary citizen to get through with a message that requires action at the top? Channels, supervisors' approvals, okays by supervisors' supervisors—the only communication the top level has is with itself; generals talk to other generals, Bureau Directors talk to generals and other Bureau Directors, the President talks to his advisors who talk only to each other. The communication barrier is complete and absolute."

"I could have told you that," said Sam. "I've been here longer than you have. But some of them may still read a newspaper now and then."

"What do you mean by that?"

"Call a news conference of the science editors and reporters of the major press services and big-city newspapers. Your reputation is big enough that they'll listen to you."

"You saw what they did to me last time!"

Sam shrugged. "Maybe you know a better way."

Jim took his seat at the console and watched the slow progress of the Prospector across the moon's surface. It was winding its way through an area of small, low crags. Ahead was a smooth, level plain. Jim determined to halt there and make the next probe.

Out of the corner of his eye he saw Hennesey moving toward them. He could think of nothing that would make the day more unpleasant than

Hennesey's presence.

The Project Director scanned the panels and the meters that showed the distance traveled by the Prospector.

"Why have you moved the machine so far?" Hennesey demanded. "You've used up valuable machine time that could have been used in additional probes. We may be approaching the end of the useful life of the Prospector very rapidly."

"I am aware of that," said Jim icily. "The stock of reagents aboard is nearly exhausted. I wanted to make at least one comparison probe at a considerable distance from our original site."

Hennesey grunted and remained silent, watching. Then, suddenly he cried out, "Look out! You fool—!"

Jim had seen it, too. At the edge of the crags was a ten-foot wide fissure spreading darkly on either side of the Prospector. The drives of the machine were upon it before he realized it was there. In fact, the crazy thought echoed in the back of his mind that it *wasn't* there an instant before.

He slammed his hand against the switches that sent out a reversing signal to the drives of the Prospector. But it was too late. The worm drives bit into nothingness as the machine toppled slowly at the edge of the crevasse. And in that moment, as the image on the television screen teetered crazily, Jim had the impression that he was looking into the black depths of utter horror. There was a blackness oozing and writhing faintly in the depths—that could have been thirty or a hundred feet deep. But he had seen just such a black horror once before.

When the EEG signals from the moon first smashed into his brain!

He glanced at Sam. Sam was staring in a kind of intense horror that told Jim he recognized it, too.

The image tilted abruptly against the black moon sky. Then the screen went dark. And Jim had the feeling that the blackness had closed over him.

But Hennesey had sensed nothing of this. He was cursing and raging beside Jim. "You blind, brainless fool! You wiped out a billion-dollar experiment because you weren't looking! You're through, Cochran! Get everything that's yours and be out of here in ten minutes!"

Hennesey whirled and strode away, his rage reeking through the atmosphere of the room.

Jim stood up and moved to the back of the panel. He opened the plastic doors and clipped the last ten feet from the spool of TV recording tape and slipped it in his pocket. When he returned to the other side of the console, Sam was waiting for him.

"Where are you going?" said Jim.

"With you."

"Where's that?"

"I don't think you know, but I do. I'll tag along and see if I'm right."

"You're crazy. Didn't you just hear Hennesey fire me?"

"Yeah. I quit at the same time."

"You're really crazy."

Jim had a few textbooks and scientific papers in his desk. He arranged for one of his men to clean them out. He didn't feel that he could endure remaining in the station any longer.

Tom Banning followed them out into the sunshine of the parking lot. "I'm sorry," he said, "but it looked as if what happened back there was rather inevitable."

"It was," said Jim. "I'd have kicked his teeth in sooner or later. It's better this way."

"What will you do now?"

"Ask Sam. He seems to think he has some crazy idea of what I'm going to do next. I sure don't."

"The news conference," said Sam. "You'd better call it right away before news of your dismissal gets out. They may think you just want to unload some sour grapes if they hear of that first."

"Yeah, I guess you're right. Will you back me up in the conference, Tom?"

The doctor nodded. "Gladly. It's pretty hard to believe, but you've got me believing."

Jim was personally acquainted with most of the newsmen who showed up for his conference. He had met them and helped them get stories on the Prospector during the past two years. They were sympathetic toward him.

He began his story by reviewing his initial discovery of the difference in moon elements. He explained the analysis and showed them samples of the telemetry record. Then he eased slowly into his discovery of fossil hydrocarbons and finally the living hydrocarbons. He watched carefully as he moved deeper into the story. He didn't want to lose them here.

They stayed with him, incredulous but confident that he knew what he was talking about. It was when he spoke of the fluctuating potential measurements, that proved to be interpretable as EEG recordings that he almost lost them. But he introduced Tom Banning quickly to verify his statements. And Tom introduced the EEG machine itself. He offered to demonstrate. A half dozen of the reporters tried it. *They* had no doubts, afterward.

"You can almost draw your own conclusions," said Jim in winding up the conference. "That thing is out there in our sky. There's no doubt about it. I've shown you what we know. Now let me tell you what I believe:

"There is some form of life in the moon. It is not merely *in* the moon. It *is* the moon. I believe its bulk occupies almost the entire volume of the moon. I believe this nemesis was spawned incalculable eons ago in a time and a space that is literally outside our own. It was driven out of that time and space by intelligent beings who could not destroy it, but who could at least exile it in a state of dormancy. Or perhaps they thought they had destroyed it and wanted not even the remains in their own domain. Perhaps the craters of the moon were caused by bombardment intended to destroy the thing.

"But it is not dead. It *was* dormant. Now, our laser probings have stirred it to feeble life. It made a deliberate effort to capture or destroy the Prospector by opening a fissure beneath it. My TV film recording proves that the fissure was not there previously.

"What are we to do about it? That is why I have called you here. Consider that the science of the intelligences in the domain that spawned this thing could not destroy it. What chance has our feeble science and powers against

such a force? Hydrogen bombs would probably serve only to feed it the energy for which it is starved.

"We must cease our lunar exploration program at once. We can hope that it is not too late. If it is not, this thing may relapse into the dormancy from which it has been shaken. We can only hope.

"But if we persist in our explorations and our probings of the moon we are certain to loose upon ourselves a living force that our entire world of science will be helpless to overcome.

"We must stop the moon program now!"

They kept him for another two hours with questions and demands for further information. He gave them everything he knew, and when they finally left, he felt that a sane and correct story of his findings would be published. He waited for whatever results would be published by the news services the following morning.

He waited.

There was nothing.

Eddie Fry called him two days later. Eddie was the reporter who knew him best. "They killed the story," said Eddie. "We had to clear it with government sources, and they persuaded every press association and newspaper that knew about it to kill it. They said it would destroy the national economy that was being built up on the space program. We tried to make them believe it, Jim, but we couldn't do it. It was hard enough to be convinced when we were listening to you. Second hand, it just wouldn't go over. You really can't blame them.

"They're doing something else, too. They're really going to nail you for this thing. A story is being released about your dismissal. It is said that you were released for fantastic and unreliable theories and for incompetence that resulted in the loss of the Prospector. I'm sorry as hell, Jim. I wish we could kill that one, but there's not a thing we can do for you."

"It's o.k., Eddie," said Jim. "I know how it is."

Crackpot. He was finished.

He called Allan at his base that night. His brother-in-law's voice was icy as he answered. "What do you want, Jim?"

"Come down over the weekend, can you, Allan? I've got something important I want to talk to you about."

"Listen, Jim. Stay away from me! Don't call; don't try to see me. Don't send me letters or telegrams. Nothing! Do you understand that?"

"What the devil—?"

"They're investigating me. Because of you. They want to know how much I've been listening to your crackpot notions. They're afraid maybe it will produce an instability that will make me unfit for the moon trip. If I lose out, it will be because of you!"

"That's what I want to talk to you about. Allan, you've got to listen to me! You won't get off the moon alive—"

The phone went dead. Jim hung up slowly and went back to the living room where Mary sat in tense, white fear. She had heard Jim's side of the conversation. She guessed what Allan had said.

"It's no use," said Jim. "Don't try to reach him. He'll hate you forever."

It was no use to run, but they ran anyway. This was what Sam meant when he said he knew what Jim was going to do. Jim Cochran was completely blackballed in his own profession. As he said, he couldn't have gotten a job stirring with a wooden paddle in a soap factory.

Tom Banning and his family went with them. They went as far north as they could and finally stopped running on the edge of the Canadian wilderness. They pooled their funds and bought some wheat land and some cattle stock and tried to stop thinking beyond the end of each day.

They were grateful for the absence of television, but they kept a radio. Through it, they learned when the Apollo finally took off with its three-man

crew. They followed its two and a half day journey through space and heard the voice of Captain Allan Wright announce they were in lunar orbit.

A few hours later the landing capsule was disengaged from the spaceship and Captain Wright and William Chambers rode it down. Their voices were heard in exultation as they announced their first steps on the surface of the moon.

It was night in northern Canada when the landing was made. Jim and Sam and Tom and their families were outside watching the full moon, trying to imagine how it was up there. From the house they heard the radio relaying the voices of the astronauts. The voices were relayed to earth through the more powerful transmitter of the orbiting Apollo, but as the spaceship circled the moon the voices of the men on the surface were lost. Then they returned once more as the ship came over their horizon.

For five orbits their voices came and went as they described their sensations and exulted in the first minutes of their achievement. Then, on the sixth orbit, there were no voices. There was only the sudden, shrill cry of the third crewman, Don Anderson, who manned the orbiting ship.

"Allan! Bill! Apollo to capsule: Come in, please. Bill—where are you—I can't even see your capsule. I'm passing right over the spot. Apollo to Base: I can't locate the capsule through the telescope. It looks like a big crevasse right where the capsule was, but it wasn't there before. Allan—Bill—Come in! Come in!"

Jim heard the sudden sob that shook Mary. He put his arm about her shoulders and led her into the house.

Don Anderson remained in lunar orbit for two more days. Then he was ordered home. He landed safely.

There were expressions of national sorrow over the unexplained loss of the two astronauts, but plans were renewed for the next voyage. The President said that sacrifices must be expected if this great goal were to be achieved, and that it would be a betrayal of those who had already given their lives if the work were to stop now.

In Canada that winter, Jim was sure the wolves howled on cold, moonlit nights more than ever before. And something new was happening to the moon. The silver light was taking on a faint tint of orange. The radio told of

a very learned report by some astronomer who spoke obscurely of changes in albedo and percentages of atmospheric dust and angstroms of sunlight. Any fool could see the moon was changing color.

Jim listened to the wolves howling in the forest, and he thought of Cramer's Pond when he was a boy, and of a machine tumbling into a crevasse where a terrible darkness lay, and he wondered how long it would be.

THE END

*** END OF THE PROJECT GUTENBERG EBOOK STAY OFF THE
MOON! ***

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